A Unified Torque-controlled Locomotion Framework for the VOLTA Humanoid Robot via IS-MPC and Passivity-based WBC

Tanawat Pawanta
Institute of Field Robotics (FIBO)
King Mongkut's University of Technology Thonburi
Bangkok, Thailand
tanawat.pawa@kmutt.ac.th

Eakkachai Pengwang
Institute of Field Robotics (FIBO)
King Mongkut's University of Technology Thonburi
Bangkok, Thailand
eakkachai.pen@kmutt.ac.th

Supachai Vongbunyong
Institute of Field Robotics (FIBO)
King Mongkut's University of Technology Thonburi
Bangkok, Thailand
supachai.von@kmutt.ac.th

Kitti Thamrongaphichartkul*
School of Mechanical and Aerospace Engineering
Nanyang Technological University (NTU)
Singapore, Singapore
kitti.t@ntu.edu.sg
*Corresponding author

***Abstract*—Maintaining stable and robust locomotion remains a fundamental challenge for humanoid robots due to their complex, high-dimensional dynamics. This paper presents a locomotion framework that integrates an Intrinsically Stable Model Predictive Control (IS-MPC) gait planner with a passivity-based Whole-Body Control (WBC) gait regulator for torque-controlled humanoids. The gait planner utilizes IS-MPC to generate stable Center of Mass (CoM) and swing foot trajectories while satisfying Zero Moment Point (ZMP) stability criteria and stability constraints. Subsequently, the WBC regulator translates these task-space trajectories into optimal joint torques by solving a constrained quadratic programming optimization problem. The proposed framework is evaluated through multibody simulations within the Drake environment. The simulation results demonstrate successful omnidirectional walking capabilities on a rigid floor. The integration effectively bridges simplified reduced-order planning with full-body torque execution, providing a solid foundation for robust humanoid balance control.**

***Index Terms*—humanoid locomotion, model predictive control, whole-body control**

I. Introduction

Humanoid robots offer superior mobility over traditional wheeled platforms in discontinuous, multi-level, and unstructured environments, making them highly suitable for high-risk disaster response or industrial maintenance tasks [1], [2]. However, maintaining robust balance during humanoid walking is inherently difficult due to their high-dimensional, underactuated floating-base dynamics and hard-to-stabilize dynamic poses [3]. Managing these underactuated floating-base dynamics requires real-time control frameworks capable of simultaneously optimizing high-level trajectory horizons and low-level torque distributions [4]. State-of-the-art architectures manage this complexity using a hierarchical cascaded scheme [4]. The high-level planning layer utilizes simplified, reduced-order models—such as the Linear Inverted Pendulum (LIP)—to execute Model Predictive Control (MPC) for continuous Center of Mass (CoM) and footstep trajectory generation [5], [6]. Downstream, a low-level regulation layer maps these task-space references into full-body joint configurations using torque-based Whole-Body Control (WBC) methods to optimize ground reaction forces [7].

Despite its theoretical benefits, integrating the MPC-WBC pipeline remains challenging. Kinematic joint-space tracking setups are prone to singular configurations and exhibit poor compliance during environmental interactions. Conversely, implementing torque-based WBC systems creates a fragile coupling layer with the reduced-order planner. Without physical feedforward inputs, arbitrary minimization of ground forces by the optimization solver introduces severe command fluctuations and high-frequency force chattering during support phase transitions [8].

To bridge these gaps, this paper introduces a unified torque-controlled locomotion framework evaluated on the VOLTA humanoid platform [9] (see Fig. 1). The architecture seamlessly integrates an online Intrinsically Stable MPC (IS-MPC) gait planner with a passivity-based torque WBC [8], [10]. Driven by user-defined omnidirectional velocity commands, the framework explicitly introduces a novel CoM-based force reference pre-calculation. This intermediate stage generates physical feedforward ground reaction force profiles that effectively suppress numerical solver chatter. The framework's robustness is thoroughly validated through Drake MultibodyPlant simulations across diverse walks on a rigid floor.

This article is organized as follows: Section II reviews related work and contributions; Section III provides a system overview; Sections IV and V formulate the IS-MPC planner

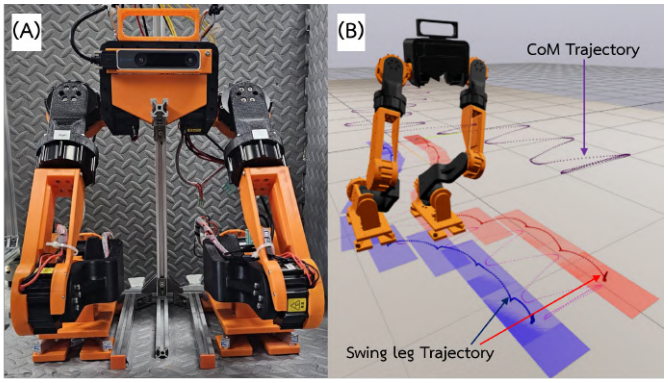


Fig. 1. Overview of the VOLTA humanoid robotics platform: The custom-built, 12-DoF (6-DoF per leg) prototype of the VOLTA robot. (B) Snapshot of the omnidirectional walking in the simulation environment.

and passivity-based WBC regulator, respectively; Section VI analyzes experimental results; and Section VII concludes the work.

II. Related Work and Contributions

A. Model Predictive Control in Humanoid Locomotion

Model Predictive Control (MPC) has become an essential framework for online trajectory optimization in humanoid locomotion, offering the capability to enforce strict system constraints over a finite prediction horizon [6]. While real-time motion generation advanced significantly with the foundational introduction of the 3D Linear Inverted Pendulum Model (LIPM) by Kajita et al. [5], earlier traditional methods still relied heavily on offline trajectory generation or analytical solutions. Consequently, these prior approaches inherently lacked the robust stability required to handle unexpected external disturbances and model uncertainties—limitations that modern online predictive frameworks actively resolve.

However, standard MPC formulations for bipedal walking frequently encounted stability and feasibility challenges over long horizons, potentially leading to diverging CoM trajectories. To resolve this, Scianca et al. introduced the concept of Intrinsically Stable MPC (IS-MPC) [10], [11]. By embedding a stability constraint directly into the optimization framework, IS-MPC guarantees bounded CoM trajectories regardless of the duration of the control horizon. This method has been expanded to handle uneven ground navigation [12] and non-convex constraints [13].

B. Whole-Body Control and Cascaded Architectures

To execute the high-level task-space trajectories generated by an MPC, WBC frameworks govern full-body dynamics by solving multi-objective Quadratic Programs (QPs). The WBC maps desired CoM and operational-space foot targets into optimal joint torques while simultaneously enforcing floating-base dynamics, actuator saturation limits, contact unilaterality, and friction cones constraint [7].

However, the cascaded integration between reduced-order planners and full multi-body regulators is highly sensitive. During contact state switches—such as foot-strike impacts and double-support weight shifts—standard geometric tracking schemes force the low-level QP solver to minimize contact wrenches arbitrarily to fulfill algebraic costs [14]. This mismatch induces severe command fluctuations, high-frequency torque oscillations, and vertical force chattering. Although passivity-based [1] formulations have been integrated into WBC to regulate internal energy dissipation [8], seamlessly suppressing force chatter across omnidirectional pathways remains an active open challenge.

C. Core Contributions

To bridge the gap between high-level bounded trajectory planning and smooth full-body force distribution, this paper presents a unified, cascaded control framework evaluated on the VOLTA humanoid robot platform. The primary contributions are summarized as follows:

1) *Integrated Architecture:* A hierarchical coupling of an IS-MPC gait planner with a passivity-based Whole-Body Control gait regulator.
2) *Force Pre-calculation:* An analytical CoM-based feed-forward model that generates ground reaction forces to eliminate numerical chatter and vertical discontinuities during double-support transitions.
3) *Omnidirectional Validation:* High-fidelity multibody dynamics validation within the Drake simulation environment demonstrating accurate tracking across diverse gait patterns on a rigid floor surface.

III. System Overview

The proposed locomotion framework (see Fig. 2) for the VOLTA humanoid robot is structured as a hierarchical, multi-rate cascaded control loop designed to decouple task-space trajectory generation from high-dimensional full-body execution.

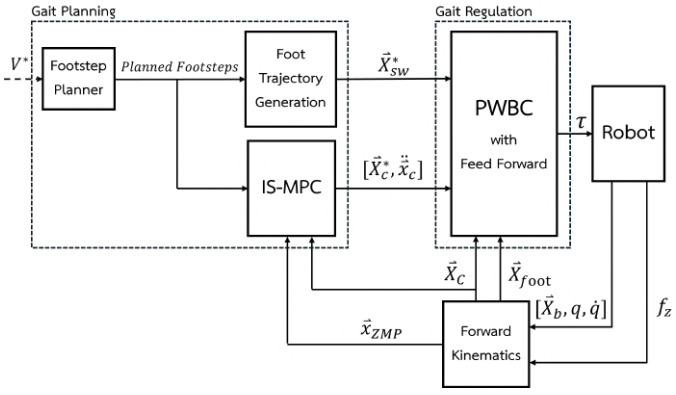


Fig. 2. Block diagram of the proposed hierarchical locomotion framework for the VOLTA humanoid robot. The architecture is split into a high-level gait planning and a low-level gait regulation.

[1] In dynamic control theory, passivity provides a mathematical framework for stable environmental interaction by ensuring the robotic system acts strictly as an energy sink or storer, rather than an unbounded energy source.

The control sequence is initiated by user commands defining a desired omnidirectional velocity vector $V^* = [v_x^*, v_y^*, \omega_z^*]^T$, where v_x^*, v_y^*, and ω_z^* denote longitudinal, lateral, and yaw target velocities, respectively. In the *Gait Planning Layer*, an IS-MPC planner utilizes a simplified, reduced-order Linear Inverted Pendulum Model (LIPM) to optimize footstep locations and CoM profiles over a finite preview horizon while adhering to ZMP stability boundaries. The resulting task-space references—the desired CoM position (X_c^*), CoM acceleration ($\ddot{X}_c^*$), and swing foot trajectory (X_{sw}^*)—are passed through the *Gait Regulation Layer* ingests the aggregated reference set ($X_c^*, \ddot{X}_c^*, X_{sw}^*, F_z^*$), where a passivity-based Whole-Body Control framework maps these targets onto the full robot configuration.

By solving a constrained, multi-objective Quadratic Program (QP), the WBC explicitly accounts for floating-base dynamics, and friction cones to output the optimal joints torque (τ). The entire framework is validated on a rigid floor surface within Drake [15] simulation environment.

IV. Gait Planning: Intrinsically-Stable MPC

This section details the high-level trajectory planning layer, which leverages an online predictive control loop to generate stable reference trajectories for the CoM and swing feet.

A. Reduced-Order Model and Pre-Defined Gaits

The horizontal relationship between the CoM and the ZMP is governed by the Linear Inverted Pendulum (LIP) dynamics:

$$\ddot{x}_c = \eta^2(x_c - x_z), \qquad \eta = \sqrt{\frac{g}{z_c}} \tag{1}$$

where x_c and x_z represent the CoM and ZMP positions, g is gravity, and z_c is the constant CoM altitude (see Fig. 3). The continuous system is discretized via Euler integration at sampling time $= 0.01$ s into a state-space model with state vector $\vec{X} = [x_c, \dot{x}_c, x_z]^T$ and control input $U = \dot{x}_z$. The lateral axis maintains an identical decoupled structure.

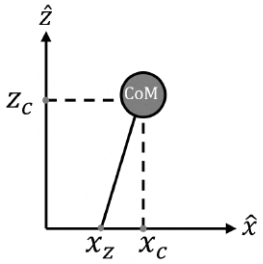


Fig. 3. Linear Inverted Pendulum (LIP)

To minimize real-time computational overhead, footstep placement configurations and orientations are pre-defined and generated offline across various reference velocity vectors V^*. This sequence is synthesized utilizing the gait generation algorithm proposed in [10]. The corresponding swing foot trajectory is modeled via a 3rd-order polynomial.

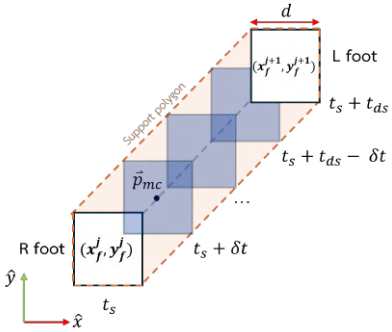


Fig. 4. Visualization of the moving ZMP constraint during the double-support phase. The admissible ZMP region is bounded within a shifting square support polygon p_{mc} of side length d, transitioning continuously over the time interval $[t_s, t_s + t_{ds}]$ from the support right foot (x_f^j, y_f^j) to the target left foot (x_f^{j+1}, y_f^{j+1}).

B. ZMP and Terminal Stability Constraints

To guarantee dynamic balance, the instantaneous ZMP must remain within a square support polygon of side length d. During the double-support phase, a moving constraint matrix linearly interpolates the bounding box center $\vec{p}_{mc}$ (see Fig. 4) between successive footsteps:

$$-\frac{d}{2} \leq \left(\vec{p}_z^{k+i} - \vec{p}_{mc}^{k+i}\right) \leq \frac{d}{2} \tag{2}$$

where $\vec{p}_z^{k+i}$ is the predicted ZMP position at the i-th look-ahead preview step.

Diagonalizing (1) yields eigenvalues $\pm\eta$ and isolates the modal coordinate $x_u = x_c + \dot{x}_c/\eta$, which propagates according to the unstable subsystem dynamics $\dot{x}_u = \eta(x_u - x_z)$. In order to prevent long-term divergence over an infinite preview horizon, an IS-MPC framework is enforced [10]. Under the periodic-tail assumption, this stability requirement reduces to a computationally efficient terminal constraint applied at the control horizon boundary C:

$$\dot{x}_u^{k+C} = \dot{x}_u^k \tag{3}$$

C. Optimization Problem Formulation

Compiling the state sequence X_i and control inputs $U_i = [\dot{x}_z, \dot{y}_z]_i^T$ over a finite preview horizon C, the online gait generation problem is formulated as a QP following the tracking objective and cost structures validated in [16]:

$$\min_{X,U} \sum_{i=0}^{C-1} \left(\|U_i\|^2 + \beta \left\|\vec{p}_z^i - \vec{p}_{mc}^i\right\|^2 \right) \tag{4}$$

$$\text{Subject to (2), (3)}, \quad |\dot{x}_z| \leq \dot{x}_z^{\max}, \quad X_0 = \vec{X}_{current}$$

The cost function minimizes the ZMP velocity input rate while penalizing tracking deviations from the moving constraint box center using a positive weighting factor β. The first optimized state step X_1 is then extracted to provide the reference task-space CoM trajectory for the low-level whole-body regulator.

V. Gait Regulation: Passivity-Based WBC

The regulation layer maps task-space references into optimal joint torques by solving a multi-objective QP at 1 kHz while satisfying floating-base dynamics and multi-contact inequality constraints.

A. Multi-Body Dynamics and Impedance Tracking

The partitioned equations of motion isolating the 6-DoF unactuated floating-base coordinates are expressed as:

$$\begin{bmatrix} M_1 \\ M_2 \end{bmatrix} \begin{bmatrix} \dot{\nu}_c^d \\ \ddot{q}^d \end{bmatrix} + \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \begin{bmatrix} \nu_c^d \\ \dot{q}^d \end{bmatrix} = \begin{bmatrix} \vec{W}_g \\ \tau \end{bmatrix} + J^T \begin{bmatrix} \vec{W}_{imp,c} \\ \vec{W}_{grf} \\ \vec{W}_{imp,sw} \end{bmatrix} \quad (5)$$

where $M_1, C_1 \in \mathbb{R}^{6\times\bar{n}}$ and $M_2, C_2 \in \mathbb{R}^{n\times\bar{n}}$ ($\bar{n} = n + 6$) represent the partitioned inertia and Coriolis matrices. The vector $\vec{W}_g = [0_{1\times5}, mg]^T$ is the gravitational wrench, τ is the joint torque vector, and J^T maps task wrenches into joint space. Translational tracking is governed by the impedance law $f_{imp} = K_p(x_d - x) + K_d(\dot{x}_d - \dot{x})$, where f_{imp}, K_p, K_d are force, positive-definite diagonal stiffness and damping matrices. To prevent kinematic singularities, torso attitude tracking utilizes unit quaternions (ϵ, δ) via $n_{imp} = -2(\delta I + \hat{\epsilon})K_r\epsilon + D_r(\omega_d - \omega)$, where n_{imp}, K_r, D_r are moment, rotational stiffness and damping matrices. The combined tracking wrench is compiled as $\vec{W}_{imp} = [n_{imp}^T, f_{imp}^T]^T$.

B. Force Pre-Calculation

To suppress numerical chattering during double-support weight shifts, an analytical vertical ground reaction force reference $F_z \in \mathbb{R}^{\alpha\times1}$ ($\alpha = 8$ contact points, as shown in Fig. 5) is calculated via a pseudo-inverse mapping of the Center of Pressure (x_P, y_P):

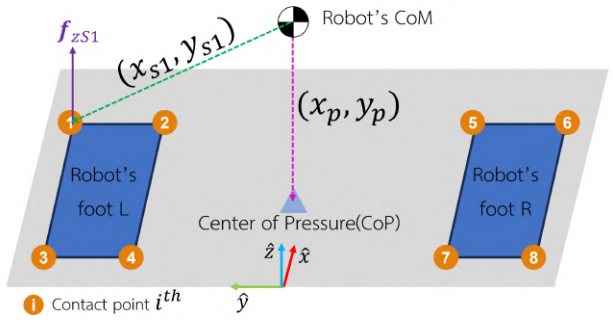


Fig. 5. Schematic of the robot's contact points during the double-support phase. The framework maps the Center of Pressure (x_P, y_P) and CoM to an analytical vertical ground reaction force reference vector $F_z \in \mathbb{R}^{\alpha\times1}$.

$$\begin{bmatrix} x_P \\ y_P \\ 1 \end{bmatrix} mg = \underbrace{\begin{bmatrix} x_{S1} & x_{S2} & \cdots & x_{S\alpha} \\ y_{S1} & y_{S2} & \cdots & y_{S\alpha} \\ 1 & 1 & \cdots & 1 \end{bmatrix}}_{A_z \in \mathbb{R}^{3\times\alpha}} \underbrace{\begin{bmatrix} f_{zS1} \\ f_{zS2} \\ \vdots \\ f_{zS\alpha} \end{bmatrix}}_{F_z \in \mathbb{R}^{\alpha\times1}} \quad (6)$$

From (6), it can be observed that the contact mapping matrix A_z is not a square matrix; thus, a unique direct algebraic inverse does not exist. Expanding on full-body compliant force distribution principles [17], this underdetermined system is resolved utilizing the Moore-Penrose pseudo-inverse $A_z^{\#}$ to compute the explicit force distribution vector $F_z^* = [f_{grf,z}^d(l), f_{grf,z}^d(r)]^T$:

$$F_z^* = A_z^{\#} \begin{bmatrix} x_P \\ y_P \\ 1 \end{bmatrix} mg \quad (7)$$

where $A_z \in \mathbb{R}^{3\times\alpha}$ contains the relative contact coordinates. Aggregating F_z^* yields the target feedforward wrench $\vec{W}_{grf}^d = [0_{1\times5}, f_{grf,z}^d(l), 0_{1\times5}, f_{grf,z}^d(r)]^T$.

C. Optimization Problem

The QP problem can be formulated as:

$$\min_{\vec{W}_{grf}} \left(\frac{1}{2}\delta_c^T Q_c \delta_c + \frac{1}{2}\delta_f^T Q_f \delta_f \right) \quad \text{s.t. } \vec{W}_{grf} \in \mathcal{FC} \quad (8)$$

where $\mathcal{FC}$ denotes constraints enforced by contact unilaterality, friction cones, and center-of-pressure constraints. The tracking residuals are defined as $\delta_c = J_{grf,c}^T \vec{W}_{grf} - M_1\dot{\nu}_c^d - C_1\nu_c^d + \vec{W}_g + \vec{W}_{imp,c} + J_{imp,c}^T \vec{W}_{imp,sw}$ and $\delta_f = \vec{W}_{grf} - \vec{W}_{grf}^d$. The final joint torque command incorporates a stance-leg to suppress localized high-frequency foot oscillations during single-support phases. A stance-leg stabilization wrench $\vec{W}_{stab} = -D\nu_{foot}$ is then incorporated following a passivity-based pure damping formulation [8], where $D \in \mathbb{R}^{6\times6}$ is a positive-definite diagonal damping matrix and ν_{foot} denotes the spatial velocity vector of the stance foot. The final joint torque vector $\tau_{actuate}$ is then mapped via the principle of virtual work:

$$\tau = \tau_g + \begin{bmatrix} J_l^T & J_r^T \end{bmatrix} \begin{bmatrix} \vec{W}_l + \vec{W}_{stab,l} \\ \vec{W}_r + \vec{W}_{stab,r} \end{bmatrix} \quad (9)$$

where τ_g compensates for gravity effects, and J_l, J_r represent the leg Jacobians relative to the torso. The foot wrenches $\vec{W}_l$ and $\vec{W}_r$ are dynamically assigned according to the active contact phase. For example, when the left leg is the stance leg, the left leg applies the optimal ground reaction wrench ($\vec{W}_l = \vec{W}_{grf}$) to stabilize the floating base, while the right leg tracks the required impedance wrench ($\vec{W}_r = \vec{W}_{imp,sw}$).

VI. Experiments and Results

The integrated hierarchical control framework is validated using multi-body dynamics simulations within the Drake environment.

A. Simulation Configuration and Parameters

The contact interface between the robot's soles and the rigid floor is modeled via a high-stiffness penalty method, with a surface stiffness coefficient of 5×10^6 N/m and a surface dissipation coefficient of 50 s/m. The high-level IS-MPC gait planner operates with a sampling time of 0.01 s over a finite control horizon of $C = 100$ steps. The target CoM

tracking altitude is fixed at $z_c = 0.35$ m. The passivity-based Whole-Body Controller executes at 1 kHz, with the output joint torques limited at a 4 Nm limit to accurately reflect the physical motor constraints of the actual hardware platform.

B. Force Distribution and Chattering Suppression Analysis

The core contribution of this framework—the CoM-based force reference pre-calculation is validated by conducting a comparative analysis against a baseline architecture. The evaluation tracks the vertical ground reaction forces (F_z) across continuous contact state transitions on the rigid floor surface.

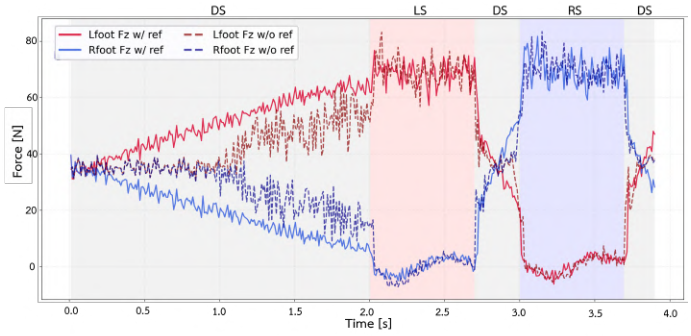


Fig. 6. Vertical ground reaction forces (F_z) during the transition from double-support to single-support phases under the baseline formulation (without pre-calculation) and the proposed framework.

Under the baseline formulation, where the target ground reaction reference wrench is set to zero ($\vec{W}^d_{grf} = 0$), the optimization variable functions purely as a force minimization term within the multi-objective QP. As a result, during the double-support (DS) weight-shifting phases, the system experiences severe numerical force chattering and high-frequency command spikes (at 1 to 2 s in Fig. 6). This behavior is a result of an algebraic conflict within the solver, which attempts to minimize contact wrenches arbitrarily while simultaneously enforcing the rigid multi-body equations of motion required to support the robot's mass.

On the contrary, the proposed CoM-based force reference pre-calculation scheme mitigates this phenomenon by introducing a physically consistent feedforward baseline prior to QP optimization. Consequently, vertical ground reaction forces transition smoothly and linearly (see Fig. 6). This eliminates joint torque discontinuities and protects the actuator hardware from high-frequency mechanical vibrations.

C. Omnidirectional Walking and Trajectory Tracking Performance

The capabilities of the integrated control framework are thoroughly evaluated across six distinct, user-commanded omnidirectional gait trajectories on the rigid floor as shown in Fig.7.

As shown across the simulation runs, the framework achieves stable locomotion under all velocity profiles. The robot successfully executes the sharp directional transitions

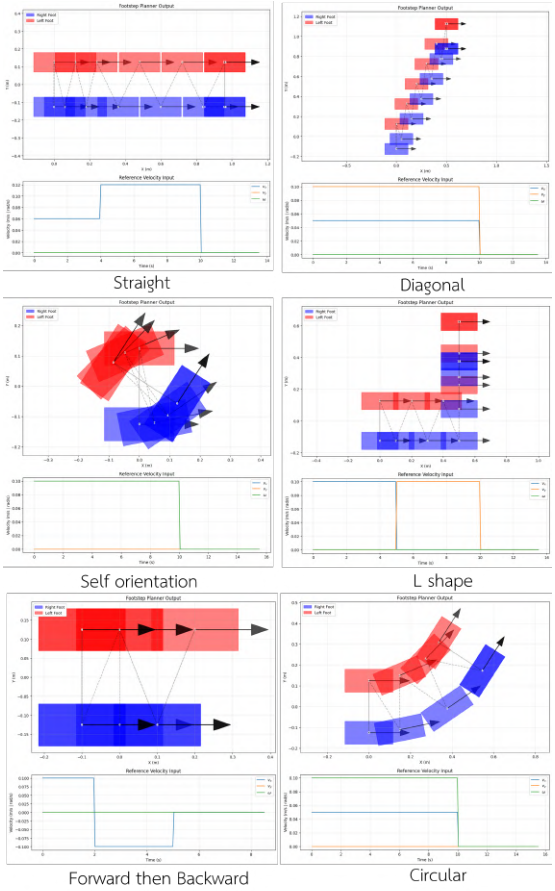


Fig. 7. Six distinct gait patterns under varying velocity profiles: Straight, Diagonal, Circular, Self-orientation, L-shape, and Forward-then-Backward locomotion. For each pattern, the top plot shows the planned footsteps (red representing Left foot, blue representing Right foot), and the bottom plot displays the corresponding reference velocity input over time.

required by the L-shape and combined paths without loss of balance or kinematic constraint violation. To evaluate the dynamic trajectory tracking capability of the integrated architecture, the tracking response of the CoM and the floating-base attitude are analyzed under the most aggressive locomotion profile (the combined walk sequence). As demonstrated in Fig. 8, the framework exhibits exceptional fidelity in tracking the high-level references generated by the IS-MPC planner.

Quantitatively, the maximum absolute CoM position tracking error along both the longitudinal and lateral axes of the horizontal walking plane is successfully bounded below 5 mm throughout the entire maneuver. Concurrently, the maximum absolute orientation tracking error remains strictly within a tight margin of 0.05 rad. This tracking precision indicates that the passivity-based WBC regulator effectively distributes joint torques to track task-space targets while preserving multi-contact constraints.

VII. Conclusion and Future Work

This work established a unified torque-controlled locomotion framework for the VOLTA humanoid by integrat-

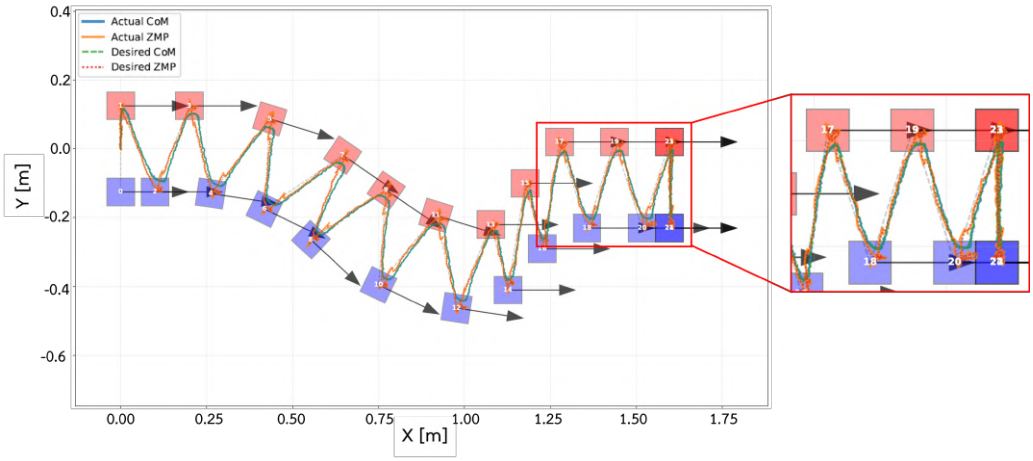


Fig. 8. CoM tracking performance and floating-base orientation errors during the continuous combined omnidirectional walking sequence on a rigid floor surface.

ing an IS-MPC gait planner with a passivity-based WBC. By implementing a novel CoM-based force reference pre-calculation layer, the architecture successfully eliminated the numerical solver chattering that typically destabilizes double-support transitions. This solution provides smooth, continuous joint torque commands that are vital for preserving actuator hardware during omnidirectional walking.

Future work will expand upon these capabilities in two primary directions:

1) **Extension to Non-Rigid Terrains:** Validation will expand to compliant surfaces using hydroelastic contact models to bridge the sim-to-real gap [18] and evaluate the regulator's robustness against terrain deformations.
2) **Physical Robot Deployment:** The control loop will be deployed on the VOLTA hardware to validate our force pre-calculation and torque distribution schemes under real-world sensor noise, and latency.

ACKNOWLEDGMENT

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